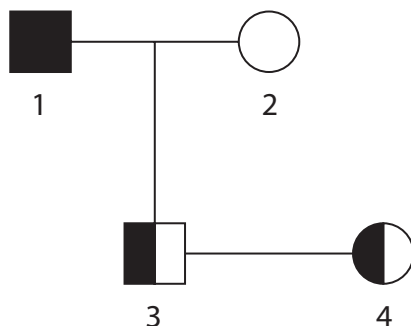


- 8 Red blood cells have a number of different groups of proteins on the cell surface membrane.

The inheritance of some of these groups of proteins is an example of codominance.

The pedigree diagram shows the blood type of individuals in part of one family. Individuals 1 and 2 are both homozygous for blood type.



Key



male with blood group M



female with blood group N



male with blood group MN



female with blood group MN

- (a) Explain why the inheritance of blood type, shown in this diagram, is an example of codominance.

(2)

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- (b) Determine the ratio of the possible blood type phenotypes of the children of individuals 3 and 4.

You must draw a genetic diagram.

(4)

Genetic diagram:

Ratio.....

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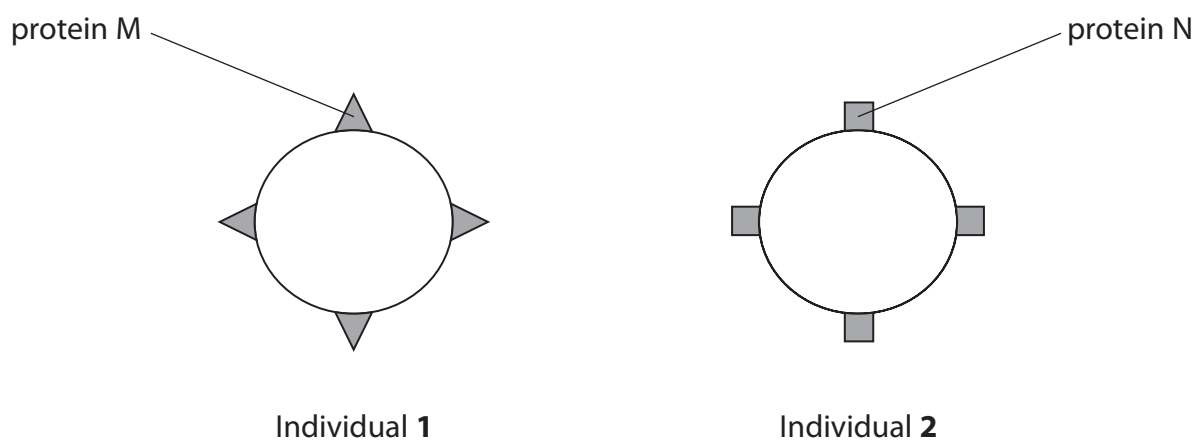
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- (c) The gene that determines this blood type codes for proteins present on the cell membranes of red blood cells.

The diagram shows red blood cells from two members of the family shown in the pedigree diagram.



- (i) Explain how the alleles present in individual 3 result in the proteins found on the surfaces of his red blood cells.

(3)

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- * (ii) Discuss the possible effects that a mutation in one of the genes coding for this blood type could have on the phenotype of individual 3.

(6)

(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 80 MARKS

